

FitBuddy – AI Fitness Plan Generator using Gemini Models

Detailed Requirement Analysis Phase

1. Introduction

The Requirement Analysis Phase defines what the FitBuddy application must provide before the development process begins. FitBuddy is planned as an AI-based fitness plan generator that uses user information and Gemini Models to produce a personalized fitness plan. This phase identifies the required inputs, system functions, user-facing features, administrative features, and quality expectations. The identified requirements will serve as the foundation for the Project Design, Project Planning, Development, and Testing phases.

2. Functional Requirements

- The system should allow users to enter relevant fitness information such as age, weight, fitness goal, and workout intensity.
- The system should receive and process the submitted user information.
- The application should prepare appropriate information for AI-based fitness-plan generation.
- Gemini Models should be used as the AI component for generating a personalized fitness plan.
- The generated fitness plan should be displayed clearly to the user.
- The application should provide suitable pages for viewing user and plan information.
- An admin dashboard should provide a structured view of user-related information.
- Users should be able to view their generated or available plan information through the application.

3. Non-Functional Requirements

- **Usability:** The application should have a simple and understandable interface so users can enter information and view results easily.
- **Reliability:** The system should process valid inputs consistently and handle unexpected input or application errors appropriately.
- **Performance:** Pages and application operations should respond within a reasonable time under normal usage.
- **Maintainability:** The application should be organized into understandable files, templates, static resources, and application logic so that future changes can be made easily.
- **Compatibility:** The web application should work correctly in a modern web browser.

4. Input Requirements

The main input requirements include user fitness details needed for generating a personalized plan. The project information identifies fields such as name, age, weight, fitness goal, and workout intensity. These values provide the information required by the application to prepare a suitable request for the AI component.

5. Processing Requirements

After receiving user information, the application should validate and organize the input before using it for fitness-plan generation. The application then prepares the required AI request and sends the relevant information to the Gemini-based component. The generated response should be processed and presented in a readable format. The application should also manage the information required by the user-facing pages and admin dashboard.

6. Output Requirements

The primary output of FitBuddy is an AI-generated fitness plan based on the information provided by the user. The output should be presented in a clear and understandable manner so that users can identify the recommended plan information easily. The application may also display user-related information and plan details through the appropriate interface or admin dashboard.

7. Technology Requirements

- Python for application development and backend logic.
- Flask-style web application structure for handling web routes and pages.
- HTML templates for creating the application interface.
- CSS/static resources for page presentation and styling.
- Gemini Models for AI-assisted fitness-plan generation.

8. User and Admin Requirements

The user side of the application should provide a straightforward method for entering fitness information and viewing the generated plan. The admin side should provide a dashboard that organizes user information in a readable format and supports access to relevant plan information. The interfaces should be designed according to the responsibilities of each type of user.

9. Expected Outcome

The outcome of the Requirement Analysis Phase is a clear and organized definition of what the FitBuddy system should accomplish. These requirements establish the scope of the application and provide measurable points that can later be used during testing. A clear requirement specification also helps ensure that the final system remains aligned with the original project objective of generating AI-assisted personalized fitness plans.

10. Conclusion

Requirement Analysis provides the foundation for the complete FitBuddy project. By defining functional requirements, non-functional requirements, inputs, processing, outputs, technology requirements, and user roles, the development team can proceed with a clear understanding of the system. The requirements identified in this phase will guide the subsequent design, planning, development, testing, documentation, and demonstration of FitBuddy.